Forecasting Boutique Fitness Studio Customer Retention

OIT 367 – Business Intelligence from Big Data

Final Project

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Note: Will you let us keep your project? Yes

Executive Summary

The last several years have seen an explosion in the number of boutique fitness studios, a trend that is disrupting the traditional incumbents in the health and wellness space. Some of the major players in the sector, FlyWheel, SoulCycle, and CrossFit, for instance, are now household names.

Following in this trend, Row House NYC ("Row House", or "the Company") is a concept gym in New York with three locations, offering a variety of cardio and strength classes centered around rowing machines. Classes can be purchased for $20-35 per session depending on the package purchased. Payment types vary and include the website, Class Pass, Gilt or free classes given by employees.

Row House is particularly interested in learning about what drives customer retention versus churn based on their class data to date. To do this, we identified two dependent variables--'power user' (attended more than 5 classes) and 'return' (took more than one class). The former helps to predict the highest value customers while the latter identifies churn at the point where it happens most, after the first visit.

We were able to access Row Houses's data via MindBodyOnline, their CRM system which logs class visits by user. We ran into some data limitations due to the fact that our data accessibility was limited to standardized reports, which limited the predictive analysis we could conduct.

Based on our analysis, we recommend that Row House focuses on four key areas to drive customer retention:

1. Introductory sales offers
2. Best practices at the Columbus Circle location
3. Strategic focus on potential new locations
4. Best practices of the trainers who most frequently teach classes

Problem Statement

Row House is a boutique fitness studio that charges on a per-class basis. Unlike other gyms, it does not have a monthly membership fee in which customers pay for 1 month (or 12 months) upfront and then have unlimited use of the facilities. Row House does not offer "open gym" usage - instead, it offers a certain number of regularly scheduled classes every day that users have to sign up in advance for, and then pay a fixed amount per class.

As a result, Row House has to win over the customer every time they make a purchase - ie, they have to reactivate the customer *every* session, rather than just once a month or once a year (for 1 month or 12 month memberships for example). The lack of a traditional monthly subscription model means that these boutique fitness studios pay extremely close attention to metrics such as customer churn and retention, and ultimately the return on investment for new user acquisition costs. To focus their marketing and operational efforts, the Row House team asked our team to focus our efforts on this topic.

Data Collection

Data Retrieval

We have cross-sectional data on ~25K athletes of Row House NYC who have taken ~72K classes. We accessed this data via their CRM system called MindBodyOnline. MindBodyOnline is a CRM designed specifically for fitness studios. The system's reports are geared mostly towards day-to-day tasks such as scheduling trainers and logging attendees rather than on deep analytics about the business.

We were limited to the standardized reports within MindBodyOnline. Due to these limitations, we reached out to the MindBodyOnline CRM vendor to ask whether we could 1) run SQL queries directly on the database so that we could pull the data we need ourselves, or 2) get a raw data dump so that we can have all the complete data and then go about scrubbing it and pulling the fields we needed. Unfortunately, the representative said no, so we had to proceed with the premade reports.

As a result, rather than class-by-class data, which would allow us the greatest insight into customer experiences, we only have data on the first class a customer attends and the most recent class they attended. In addition, Row House does not collect extensive demographic data on their customers, so the only information we have is the zip code in which they live. Further, the format of these reports is not conducive to the analysis we are undertaking. We explored the ~25 standard reports to identify where beneficial data resided and could only rely on data that included a unique identifier about the customer that we could use (e.g., e-mail address or user ID).

One of the main limitations of this data is that retention within this business is tied to the experience across each class a customer attends. However, we do not get a view into this rich experience due to the way the MindBodyOnline reports are created.

Data Preparation and Cleaning

We had to do a significant amount of data cleaning to join unique entries across all the separate tables from the raw data that was downloaded, which resulted in data entries being dropped if certain data fields were not complete. As a result, our ultimate data set ended up being reduced from ~25K customers to ~7K.

We queried the MindBodyOnline CRM system for all entries between 1/1/2014 through 3/1/2016. There were no classes recorded before 1/1/2014. Below is a table of the reports that we downloaded from MindBodyOnline, along with all the rawdata that was included in each report.

Mailing list	Name, user ID, e-mail, address (including zip code), phone number
Attendance	Name, user ID, total visits, Paid Visits, Signed-In Visits, No-Shows, Cancelled
First visit	Name, first visit date, purchase type, trainer, phone number, e-mail
No shows	Name, user ID, number of no shows/late cancels, phone number, e-mail
No return	Name, last visit date, address, phone number, e-mail

To clean this raw data, we took the following steps:

- From the Mailing List report, obtain all the Row House clients. Clean the following entries from the data:
    - Delete staff entries (specified as staff).
    - Delete any entries without an email address (because that is one of the unique identifiers that we use to join the data to other tables).
- Data from Attendance:
    - Match user ID from Mailing List to user ID on Attendance, and join the tables on this parameter, leaving # of Paid Visits, Signed-In Visits, No-Shows, Cancelled.
- Data from First Visit
    - Match email address from Mailing List to email address on First Visit, and join the tables on this parameter, leaving date of First Visit, Trainer, Purchase Type.
    - Discard the column with total # of visits b/c it doesn't match data from Attendance.
- Data from No Shows
    - Match user ID from Mailing List to user ID on No Shows.
    - The data was not comprehensive as Attendance (missing entries), so we went with the entries in Attendance for the final data.
- Data from No Returns
    - Match email address for Mailing List to email address on No Returns, and join the tables on this parameter, which provides the Last Visit Date.
- Clean up Master Data
    - Remove all N/A's (there are sometimes fields with N/A's that would make the regression analysis lose meaning, such as N/A on number of visit).
    - Handle rows with 0 for zip code; clean and change zip code into 5 digits if any are found with 4 digits.
    - For purchase type - create new categorical variables on first class type (ClassPass, GILT, free class) or Location (Chelsea, Upper East Side).

Preliminary Analysis

Trend Spotting

The primary variables that could be extracted from the MindBodyOnline data set were as follows:

Variable name	**Description**
First	First name of customer
Last	Last name of customer
User ID	ID assigned by CRM system to customer
Zip Code	Home zip code of customer
Email	Email address of customer
First visit	Date of first visit
Location	Row House location of first visit
Trainer	Trainer of first class customer attended
Type	Payment type for first visit (e.g., Class Pass, Gilt, regular, free class)
Paid visit	Number of times the customer has paid for a class at Row House
Signed in visit	Number of times the customer has showed up to class at Row House
Phone number	Phone number of the customer

Last visit	Date of last visit

Using the variables provided we created several new variables based on our opinion of important factors that may drive customer retention in order to begin testing correlations. We started with the 'First visit' date variable to see if there were any patterns corresponding to when the customer went to their first class. We then created binary variables for each of the three Row House locations, 'Columbus', 'Chelsea' and 'UES' using the 'Location' variable. This would enable us to test any patterns that were specific to a certain location. Likewise, we created binary variables for each of the four payment types from the 'Type' variable: 'Regular', 'classPass', 'GILT', and 'freeClass'. This binary variable indicated whether or not a customer lives in the same zip code as a studio. Using both the 'Paid visit' and 'Signed in visit' variables we created a 'no show' variable of the number of times a customer paid for a class, but did not show up.

Using these variables we began to run simple regressions to see what led to a retained customer. We began by running regressions using the 'Signed in visit' as the response variable. However, as we thought it through more we realized that this was skewing the data for several reasons. First, the customers who first went to Row House in the last few weeks would be expected to have a lower number of visits because they have not have as much time to return to Row House over and over again. Likewise, customers whose first visit was two years ago should on average have more visits. Additionally, promotions such as ClassPass have not been around since the start of Row House and not all of the locations have been open for the same amount of time.

We realized that we needed a way to look at what really mattered to the client, which was seeing what factors lead to customer retention. We considered two alternative dependent variables to test. The first option was average visits per week, which would account for the length of time that a customer had passed since a customer's first visit. We created a variable 'Days since first visit' to count the days from the first visit to the date the data was pulled by using the 'First visit' variable. Then we divided the 'Signed in visit' variable by the 'Days since first visit' divided by 7 to get an average visits per week since first visiting.

The second option was to consider what number of visits meant a customer was 'retained'. We decided that a reasonable cut off for return visits is looking at customers whose first visit was prior to 2016. This would give them 3 months for a return visit. We also excluded all customers who paid, but never signed in. We then created a pivot table in Excel of the percent of customers by the number of 'Signed in visits'. We learned from this that 43% of customers only have come once and 16% only have come twice. We cut this pivot table using a variety of independent variables to do preliminary analysis on which variables had the most dramatic impact on customer retention over time. see which ones Given moving from one visit to any number of following visits was such a big drop off, we decided to concentrate on using whether or not the customer signed in more than once as the response variable. We created a binary variable called 'retained' to measure or not a customer returned after their first visit.

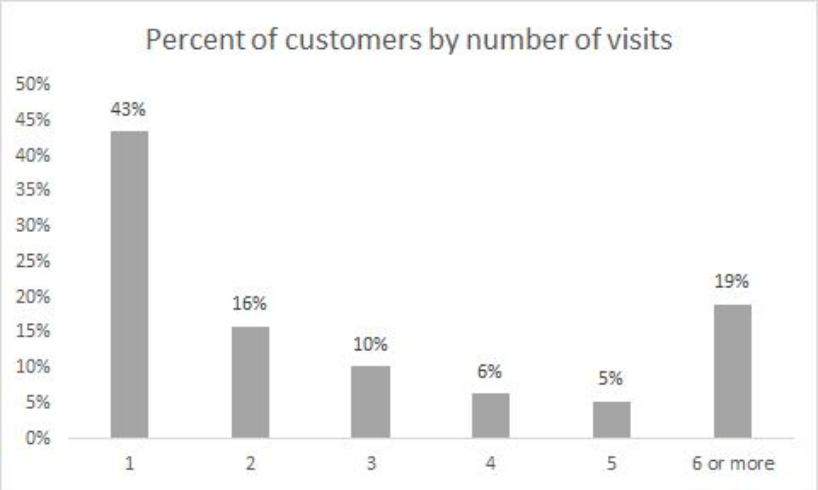

Table 1. Percent of customers by number of visits (excludes customers whose first visit was in 2016)

Year	(Multiple Items)

% of customers	Location			
Number of visits	Chelsea	Columbus	UES	Grand Total
1	52%	29%	43%	43%
2	16%	14%	18%	16%
3-5	21%	23%	21%	22%
6-10	8%	16%	13%	12%
11 or more	2%	17%	6%	7%

Year	(Multiple Items)

% of customers	Payment type				
Number of visits	classPass	freeClass	GILT	Regular	Grand Total
1	48%	43%	15%	35%	43%
2	17%	11%	9%	14%	16%
3-5	21%	13%	46%	21%	22%
6-10	11%	12%	19%	13%	12%
11 or more	4%	21%	12%	17%	7%

Table 2. Pivot tables of number of customer visits by Row House location and payment type (excludes customers whose first visit was in 2016)

Predictive Model

Data Limitations

Given the limitations of the data, we tried to be thoughtful about how we might set out building a predictive model. We initially set out to use the data on hand to see if we might be able to forecast customer churn. We defined a customer as having been retained if both 1) they had attended at least three classes and 2) they had attended a class in the previous month.

The initial results were not particularly compelling, so we considered what we had extracted from the data so far. As discussed above, we saw that the largest drop off in users were those who only went once and never returned for a second time. After a customer came back for a second class, they were more likely than not to come back for a third, and so on.

Importantly, we revisited the data we had been provided. As previously discussed, the unfortunate limitation of the data was in that we really only had significant information about a user's first class experience. That is to say, after they attend their first class, regardless of how many additional classes they attend, we do not know their unique spending or attendance habits. This makes predicting whether a user continues to come back after their second, or third, or fourth classes a slightly meaningless exercise, given that we would have been overstating the importance of the user's first visit on whether they would continue to show up.

Given all of this, we decided to further trim our data set. We threw out all users who failed to attend a class, and all users who attended a first class after January 31st, 2016. This gave us a dataset of approximately 6,330 users, all of whom had attended at least one class. We then defined a new variable called 'Return', which returned a binary response of '1' for having attended more than one class, and a '0' if they failed to return for a second class. Since we had reasonably detailed information about a customer's first experience with Row House, we figured we may be able to build a model that predicted whether or not a user would return for a second class. The importance of identifying such users is extremely important - in our revised data set, customers who attend a second class have a 70% likelihood to attend a third.

Acknowledging the limitations in trying to forecast the explanatory power of the characteristics of a user's first experience on predicting behavior beyond attending one class, we also created a binary variable called 'powerUser' that returned a '1' if a user had attended more than five classes, and a '0' if a user had not.

Data Preparation

In addition to the further trimming of the dataset that we performed, we also introduced several new variables and additional datapoints that we thought may have explanatory and predictive power. Firstly and most importantly, we added a numeric variable called 'firstDay' that assigned a numeric value to the day of the first visit. This was an extremely important addition - obviously a customer who attended their first class earlier would have had more time to attend subsequent classes. Additionally, we took the date of the first visit and extracted two variables that we defined as 'firstWeekday' and 'firstMonth' that stored the day of the week and the month of the first visit, trying to understand whether when a customer first visited could tell us something about their retention rate.

In an attempt to make use of the zip code data, we took the data and sorted it by the most frequently occurring zip codes. We then took the top thirty occurrences and replaced all of the other zipcodes with a character string, 'Other'. We also downloaded all of the Manhattan Zip Codes and created a binary variable for whether or not a customer was located in Manhattan. Further, we created another binary variable called 'Nearby' that returned a '1' if the customer was located in the same zip code as one of the three Row House studios (UES: 10036, Chelsea: 10011, Columbus Circle: 10023 & 10019), in an attempt to see whether living in near proximity would have significant explanatory power.

Similarly to the zip code, we took the Trainer data and took the bottom trainers by number of first visits with each trainer, and grouped them into one 'Other' category. The idea behind this was to remove any biases that might come about as a result of small sample sizes for those trainers who did not have a tremendous number of first time customers.

Within RStudio, the statistical software we chose to utilize for analyzing the dataset, we then binned the categorical variables 'Studio', 'Payment', 'zipCode', 'Trainer', 'firstWeekday', and 'firstMonth' into separate binary variables that we could use in a regression. From there, we created additional interactive variables between location of first visit and the corresponding zip codes for those locations - that is to say, the interactive variable "Chelsea" returns '1' if the user lives in the zip code 10011 and their first visit was to the Chelsea location. While it might be reasonable to assume that anyone who lives in 10011 would attend the Chelsea location for their first class, this helps control for users who may have differing exercise patterns and who may choose to exercise by their office instead of by

their home, especially given the lack of commitment in purchasing single classes as opposed to a monthly recurring gym membership. Additionally, it could account for customers who attended Row House prior to a location in their home zip code opening.

Model Building

Given that our dependent variables are dummy variables, we chose to use a logistic regression model rather than a simple linear model. While this means that the output regression coefficients are inherently less intuitive than those of a simple linear model, a logistic regression allows us to make use of more relevant measures of predictive power than the least squares method, which breaks down with a binary dependent variable in a linear model.

We divided the data into a training and a validation set and used a five-fold cross validation to analyze the predictive power of our models. Instead of using a least-squares analysis to measure the predictive power, we examined the average AUC from the cross-validation. At first, we ran a regression against all of the available variables. However, due to the large number of dummy variables introduced by binning 'zipCode' and 'Trainer' we decided to eliminate the variables within those two categories that did not seem to have predictive power, simplifying our model and reducing the risk of overfitting. We used the same approach for 'firstMonth' and 'firstWeekday' as well.

For each elimination, we considered the effects on predicting both 'Return' and 'powerUser'.

After finding what we deemed to be the best logistic regression models we could fit, we also tested the their predictive power against a GBM model. Using cross-validation, we did not find that the GBM model provided any significant increase in predictive power over our logistic regression model.

Summary of Results

Our logistic regression model, summarized in detail in the appendices, resulted in some interesting takeaways. For predicting 'Return', we generated an AUC of approximately 0.656, and for predicting 'powerUser', we generated an AUC of approximately 0.747.

Both models seemingly have some predictive power. Before we offer those results though, we provide one word of caution - the majority of the predictive power increase between 'Return' and 'powerUser' is likely attributable to the explanatory power of 'firstDay', or the numeric variable of the date (discussed below). That said, there are variables that have more explanatory power for one or the other dependent variables, suggesting that we may actually be able to tell with more certainty whether or not a customer will become a power user versus simply a return user.

- As discussed, 'firstDay' (numeric variable of the date) has extremely high predictive power. This intuitively makes sense - the longer you've been a customer the more likely it is that you've gone 5 times. A little more surprising is the fact that it also holds for 'Return' as well. One might figure that if you went six months ago and you haven't gone a second time having an extra month wouldn't make much of a difference.
- An interesting insight is that trainers generally do not have much predictive power. However, as mentioned above, we grouped the bottom 10 or so trainers based on number of people who they had as first class (in an attempt to limit small sample size effect) into a categorical variable called 'TrainerOther'. It turns out, of all of the Trainer categorical variables introduced, this is one of the few trainer variables with explanatory power, and it turns out to

have a significantly negative effect on a person's likelihood of returning or becoming a power user. What is surprising about this result is that a seemingly random grouping of trainers who just so happened to be the least represented on the list in terms of instances whereby they were a user's first class happened to have explanatory power for whether a customer was likely to return. We considered whether Row House diligently tracks first class retention and recognized that the trainers included in this grouping had been identified and released, explaining why they did not continue to show up more frequently in the dataset. Another potential explanation might be that these trainers teach the longer or more specialized classes that most first time users wouldn't go to, implying that the customers who take these classes first may be scared off and less likely to return. Alternatively, these could be new trainers who haven't taught many classes and who perhaps as a result are not yet as skilled. Without access to additional information, it is difficult to draw definitive conclusions.

- GILT has significant predictive power for both 'Return' and 'powerUser'. This is unsurprising for 'Return' - customers buy the GILT offering for 5/10/15 class packages at a discount. But 'powerUser' is a little different. While this could be accounting for the fact that people who bought GILT packages could've bought more than 5 class packages, it feels unlikely that the majority would. It seems that customers who purchase a package and go several times are more likely to come back after their package has expired. This is in keeping with the idea that after you have attended a second class, you are more likely than not to take a third, and so-on.
- Regular purchasers (e.g., no special offer) are not statistically significant predictors of whether they'll return or become a power user, but customers who had a free class are slightly more likely to become power users, although it does not predict with any significance simply whether a user will return or not. This could be biased by the relatively small number of instances where users in our sample received their first class free.
- Customers who take their first class on Sunday, Monday and Tuesday are also more likely to return and become a power user, particularly for Monday. Maybe one way to rationalize this phenomenon is that Monday is not a particularly popular workout day - working out Monday could be a signalling effect for being a hardcore consumer of fitness classes and makes them more likely to be the type of person who would continue coming back. However, examining our provided data, Monday is actually the most common day of the week for a user to attend a their first class.
- If you took your first class at the Columbus Circle location, you're more likely to come back for another and to become a power user.
- If you took your first class at the Columbus Circle location **and** live in Columbus Circle, you are more likely to return, but not any more likely to become a power user. This holds for the Chelsea location as well, and the 'Chelsea' variable is also significant for predicting whether you are likely to become a power user.
- Additionally, if you live in the Columbus Circle area and did not take your first class at the Columbus Circle location, it is statistically significant that you are less likely to return. There are not any significant predictive measures for similar variables for the Chelsea and UES locations.
- If you had your first class in September or October, you are more likely to become a power user, at a high level of statistical significance. There is no similar statistical effect for whether

or not you return for a second visit. This may be as a result of some sort of special promotion ran during these months, or some other factor. It seems worth examining further.

Implications for the Row House team

Based on our results, the Row House team should focus in four areas to improve customer retention:

1. **Introductory offers.** Our findings regarding the benefits of the GILT deals for the 5/10/15 class packs indicates that Row House should consider the introductory benefits provided. For example, Row House could offer new customers a 5-class pack at a very steep discount knowing that this could lead to the creation of a "power user" from whom they could receive a high customer lifetime value in terms of revenue generation. As with any significant discounting measure, the team should track the performance of this offer real-time to ensure it is having the desired effect. Similarly, free class offerings seemed to predict customers who would return, so the Row House team could experiment further with that offer as well.
2. **Columbus Circle best practices.** The Columbus Circle location is driving the most return behavior and power user behavior. As such, the Row House team should undergo a review of the Columbus Circle location's practices and determine if there are any that could be shared with other existing locations or future new locations.
3. **New locations.** The convenience factor of having a Row House gym nearby to a customer's home seems to drive repeat behavior. As the team looks to open new locations, they should consider either a) locations where they have a significant customer base they would like to retain or b) the option to open multiple small locations vs. fewer larger locations if the economics works out because of the value of locations near where their customers live.
4. **Trainers.** Use what they know about the trainers grouped in the other group--Jennie, Travis, Trevor, Steven, Brandon, Isake, Carissa and Esther--to determine the characteristics these trainers might have in common. For example, if they are all newer trainers, consider how they could implement a training program to help new trainers to be more effective from the start.

Additionally, we strongly recommend that the Row House team either hires an in-house CRM resource or contracts with a third party to have part-time support in this area. With this support, they could customize the data they can receive via APIs into the MindBodyOnline system. Reviewing the full breadth of the data they have collected to date and combining it with demographic data available for purchase could yield even richer insights into their customer behavior and specifically how to drive customer retention and pursue the best expansion strategy.

Appendix

Logistic Regression of 'Return' Against Select Variables

```
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-2.4201  -1.1174   0.5365   1.1228   1.8258

Coefficients:
                       Estimate Std. Error z value Pr(>|z|)
(Intercept)           1.312e+02  1.438e+01   9.121  < 2e-16 ***
UES                   2.471e-01  1.426e+00   0.173 0.862439
Chelsea               1.653e-01  1.269e-01   1.302 0.192816
Columbus              1.735e+00  4.462e-01   3.887 0.000101 ***
zipCode10022         -1.912e-01  2.696e-01  -0.709 0.478270
zipCode10003         -4.834e-01  1.649e-01  -2.931 0.003374 **
zipCode10005         -3.296e-01  3.279e-01  -1.005 0.314830
zipCode10014         -2.207e-01  1.798e-01  -1.228 0.219621
zipCode10013         -3.366e-01  2.773e-01  -1.214 0.224760
zipCode10019         -1.331e+00  4.396e-01  -3.028 0.002462 **
zipCode10009         -6.009e-01  1.838e-01  -3.269 0.001078 **
zipCode10128          3.402e-01  1.477e-01   2.303 0.021300 *
zipCode10016         -1.507e-01  1.571e-01  -0.959 0.337540
zipCode10023         -1.033e+00  4.385e-01  -2.356 0.018468 *
zipCode10028          1.478e-01  1.765e-01   0.837 0.402471
TrainerOther         -2.082e-01  2.137e-01  -0.974 0.329984
TrainerBethany        5.368e-01  2.410e-01   2.227 0.025928 *
firstMonthMay        -6.052e-02  1.530e-01  -0.395 0.692504
firstMonthJune        1.460e-01  1.514e-01   0.964 0.334962
firstMonthJuly        4.304e-03  1.455e-01   0.030 0.976404
firstMonthAugust      6.198e-01  2.321e-01   2.671 0.007571 **
firstMonthSeptember   2.097e-01  1.703e-01   1.231 0.218326
firstMonthOctober     1.512e-01  8.161e-02   1.853 0.063892 .
firstMonthNovember    1.525e-01  8.870e-02   1.719 0.085565 .
firstWeekdaySunday    2.106e-01  1.239e-01   1.699 0.089251 .
firstWeekdayMonday    2.542e-01  1.188e-01   2.140 0.032352 *
firstWeekdayTuesday   2.170e-01  1.168e-01   1.859 0.063033 .
firstWeekdayWednesday 5.818e-02  1.194e-01   0.487 0.626153
firstWeekdayThursday  1.775e-01  1.251e-01   1.419 0.155966
firstWeekdaySaturday  1.640e-01  1.229e-01   1.335 0.182000
StudioUES            -1.893e-01  1.110e-01  -1.705 0.088240 .
StudioColumbus        1.716e-01  1.134e-01   1.513 0.130291
PaymentRegular        2.600e-02  8.713e-02   0.298 0.765381
PaymentGILT           1.435e+00  1.916e-01   7.492 6.76e-14 ***
PaymentfreeClass     -3.570e-03  2.675e-01  -0.013 0.989351
Manhattan             2.701e-01  7.898e-02   3.420 0.000627 ***
firstDay             -3.112e-03  3.394e-04  -9.167  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 7000.0  on 5063  degrees of freedom
Residual deviance: 6533.6  on 5027  degrees of freedom
AIC: 6607.6

Number of Fisher Scoring iterations: 4
```

Logistic Regression of 'powerUser' Against Select Variables

```
Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.8794  -0.5994  -0.4107  -0.2658   2.8122

Coefficients:
                        Estimate Std. Error z value Pr(>|z|)
(Intercept)            1.948e+02  1.552e+01  12.553  < 2e-16 ***
UES                    2.229e+00  1.481e+00   1.505 0.132319
Chelsea                4.137e-01  1.908e-01   2.169 0.030120 *
Columbus               5.234e-01  6.420e-01   0.815 0.414940
zipCode10022          -1.037e+00  4.923e-01  -2.106 0.035230 *
zipCode10003          -7.687e-01  3.342e-01  -2.300 0.021431 *
zipCode10005           1.559e-01  5.022e-01   0.310 0.756268
zipCode10014          -9.827e-01  3.843e-01  -2.557 0.010560 *
zipCode10013          -4.294e-01  4.996e-01  -0.859 0.390097
zipCode10019          -2.809e-02  6.307e-01  -0.045 0.964479
zipCode10009          -5.539e-01  3.364e-01  -1.646 0.099663 .
zipCode10128           9.633e-03  1.935e-01   0.050 0.960301
zipCode10016          -5.397e-01  2.730e-01  -1.977 0.048040 *
zipCode10023          -1.759e-01  6.471e-01  -0.272 0.785739
zipCode10028           2.073e-01  2.166e-01   0.957 0.338673
TrainerOther          -4.983e-01  2.434e-01  -2.047 0.040661 *
TrainerBethany        -1.556e-01  3.909e-01  -0.398 0.690694
firstMonthMay          3.011e-01  1.690e-01   1.782 0.074819 .
firstMonthJune         3.947e-02  1.824e-01   0.216 0.828624
firstMonthJuly         5.684e-01  1.695e-01   3.353 0.000798 ***
firstMonthAugust       5.550e-01  2.359e-01   2.352 0.018653 *
firstMonthSeptember    9.412e-01  2.007e-01   4.689 2.75e-06 ***
firstMonthOctober      8.064e-01  1.312e-01   6.145 8.00e-10 ***
firstMonthNovember     5.575e-01  1.450e-01   3.844 0.000121 ***
firstWeekdaySunday     4.812e-01  1.835e-01   2.622 0.008747 **
firstWeekdayMonday     4.731e-01  1.791e-01   2.642 0.008243 **
firstWeekdayTuesday    3.931e-01  1.773e-01   2.218 0.026588 *
firstWeekdayWednesday  3.864e-01  1.820e-01   2.123 0.033717 *
firstWeekdayThursday   1.638e-01  1.968e-01   0.833 0.405100
firstWeekdaySaturday   2.863e-01  1.848e-01   1.549 0.121267
StudioUES              2.582e-01  1.639e-01   1.575 0.115179
StudioColumbus         6.824e-01  1.565e-01   4.360 1.30e-05 ***
PaymentRegular         1.610e-01  1.140e-01   1.412 0.157956
PaymentGILT            8.081e-01  1.728e-01   4.677 2.92e-06 ***
PaymentfreeClass       5.931e-01  2.938e-01   2.019 0.043511 *
Manhattan              1.885e-01  1.143e-01   1.648 0.099348 .
firstDay              -4.682e-03  3.667e-04 -12.766  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 4458.8  on 5063  degrees of freedom
Residual deviance: 3757.3  on 5027  degrees of freedom
AIC: 3831.3

Number of Fisher Scoring iterations: 5
```